EXAMINATION ANSWER BOOK

Course Code		Exam Date	
Student ID		Student Name	

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Declaration of Academic Integrity

I confirm that I have answered the questions using only materials specifically approved for use in this examination, that all the answers are my own work, and that I have not received any assistance during the examination.

Student's Signature: ______________________________

AIAA2711 (L01) - Mathematics for AI

Final Examination

Spring 2026

The Hong Kong University of Science and Technology (Guangzhou)

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Part I: True (T) or False (F) Questions

Please complete the table below with your answers to the true or false questions. Use "T" for true and "F" for false. (20%; each for 2%)

1	2	3	4	5	6	7	8	9	10

1. The Kullback–Leibler divergence $KL(P, Q)$ is symmetric, i.e., $KL(P, Q) = KL(Q, P)$ for any probability distributions P and Q.

2. For any square matrix $A \in \mathbb{R}^{n \times n}$, the trace of A equals the sum of its eigenvalues.

3. Backpropagation is a special case of forward-mode automatic differentiation.

4. Every generating set of a vector space is also a basis of that space.

5. If (X, Y) follows a bivariate Gaussian distribution, then the marginal distribution of X is Gaussian.

6. The set of integers $\mathbb{Z}$ forms a group under ordinary multiplication.

7. For any real matrix A, the non-zero singular values of A are exactly the square roots of the non-zero eigenvalues of both $A^T A$ and AA^T.

8. For any two probability distributions P and Q defined on the same sample space, the cross-entropy $H(P, Q)$ is always greater than or equal to the entropy $H(P)$.

9. The Stochastic Gradient Descent (SGD) algorithm computes the exact gradient of the loss function with respect to the current parameters in each iteration.

10. The determinant of an orthogonal matrix must be equal to 1.

Part II: Short answer questions.

(20%; each for 4%; ONLY THE ANSWERS ARE REQUIRED)

1. The singular values of a matrix $A \in \mathbb{R}^{3\times 3}$ are $\sigma_1 = 5$, $\sigma_2 = 2$, $\sigma_3 = 0.1$. If $\widehat{A}$ is the best rank-2 approximation of A (in the spectral norm), then $\|A - \widehat{A}\|_2 =$ ______________.

2. Let $\mathbf{x} \in \mathbb{R}^3$, and let $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{A}\mathbf{x}$ with $\mathbf{A} = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$. Then the gradient of f with respect to $\mathbf{x}$ is $\dfrac{\partial f}{\partial \mathbf{x}} =$ ________________________.

3. Consider a univariate linear regression model $y = wx + b$ trained using the mean squared error (MSE) loss function $L(w, b) = \frac{1}{N}\sum_{i=1}^{N}(y_i - (wx_i + b))^2$. We have two training examples: $(x_1 = 1, y_1 = 3)$ and $(x_2 = 2, y_2 = 5)$. Starting with initial parameters $w_0 = 0$ and $b_0 = 0$, if we perform one step of batch gradient descent with learning rate $\eta = 0.1$, the updated weight w_1 is ____________ and the updated bias b_1 is ____________.

4. Let X and Y be discrete random variables with the following joint probability distribution:

$$P(X{=}0, Y{=}0) = \frac{1}{4},\ P(X{=}0, Y{=}1) = \frac{1}{4},\ P(X{=}1, Y{=}0) = \frac{1}{8},\ P(X{=}1, Y{=}1) = \frac{3}{8}.$$

The mutual information $I(X; Y)$ is ____________. (in nats, using the natural logarithm). (Round your answer to three decimal places.)

5. A binary image classifier outputs predicted probabilities Q, where $Q(\text{cat}) = 0.8$ and $Q(\text{dog}) = 0.2$. The true distribution P of the test set has an equal number of cat and dog images ($P(\text{cat}) = P(\text{dog}) = 0.5$). The Kullback-Leibler divergence $KL(P \parallel Q)$ is ______________ (in bits, Round your answer to three decimal places).

Part III: Long Answer Questions.

(60%; each for 10%; PLEASE PROVIDE DETAILS FOR SOLVING THE QUESTIONS)

Question 1

Consider the affine subspace $L = \mathbf{x}_0 + U$ of $\mathbb{R}^3$, where

$$\mathbf{x}_0 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \qquad U = \operatorname{span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

Let $\mathbf{v} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$.

(a) Find the orthogonal projection $\pi_L(\mathbf{v})$ of $\mathbf{v}$ onto L.

(b) Compute the distance from $\mathbf{v}$ to L.

Question 2

Consider an unconstrained optimization problem:

$$\min_{x \in \mathbb{R}^n} f(x)$$

where $f : \mathbb{R}^n \to \mathbb{R}$ is a continuously differentiable convex function.

(a) (6 points) Let $f : \mathbb{R}^n \to \mathbb{R}$ be a continuously differentiable convex function. Prove that any local minimum of f is also a global minimum. (Hints: x^* is a local minimum of f implies that there exists $\epsilon > 0$ such that for all x with $\|x - x^*\| < \epsilon$, $f(x) \geq f(x^*)$)

(b) (4 points) Consider the following unconstrained optimization problem:

$$\min_{w \in \mathbb{R}^2} f(w) = \frac{1}{2} w^T \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix} w + \begin{pmatrix} 2 \\ 1 \end{pmatrix}^T w$$

Starting from the initial point $w_0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, perform two iterations of gradient descent with a fixed learning rate $\eta = 0.2$. Write down the values of w_1 and w_2.

Question 3

A machine learning dataset contains samples of two discrete random variables X (feature) and Y (label). Their joint probability distribution is given in the table below:

$X \setminus Y$	-1	0	1
-1	0.1	0.2	0.1
1	0.2	0.1	0.3

(a) (2 points) Compute the marginal probability distributions $P(X)$ and $P(Y)$.

(b) (2 points) Compute the conditional probability $P(Y = 1|X = 1)$.

(c) (4 points) Compute the expected values $E[X]$, $E[Y]$ and $E[XY]$.

(d) (2 points) Are X and Y independent random variables? Justify your answer.

Question 4

Let $A \in \mathbb{R}^{3\times 3}$ be the symmetric matrix

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

Prove that the function $\langle \cdot, \cdot \rangle_A : \mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}$ defined by

$$\langle \mathbf{x}, \mathbf{y} \rangle_A = \mathbf{x}^\top A \mathbf{y}, \qquad \mathbf{x}, \mathbf{y} \in \mathbb{R}^3$$

is an inner product on $\mathbb{R}^3$. Conclude that $\|\mathbf{x}\|_A = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle_A}$ defines a norm on $\mathbb{R}^3$.

Question 5

Consider the following function, which represents a single neuron with a sigmoid activation:

$$f(x) = \sigma(x^2 + x)$$

with the sigmoid function defined as $\sigma(z) = \frac{1}{1+e^{-z}}$. We will compute the gradient $\frac{df}{dx}$ at the point $x = 2$ using reverse-mode automatic differentiation.

(a) (3 points) Draw the computational graph for this function, labeling all intermediate variables.

(b) (3 points) Perform a forward pass through the graph to compute the value of all intermediate variables and the final output $f(2)$.

(c) (4 points) Perform a backward pass through the graph to compute the gradients $\frac{df}{dx}$ at $x = 2$.

Question 6

Consider the subspace $U \subseteq \mathbb{R}^4$ defined by

$$U = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \in \mathbb{R}^4 \;\middle|\; x_1 + x_2 + x_3 + x_4 = 0, \quad x_1 - x_2 + x_3 - x_4 = 0 \right\}.$$

- Find a basis for U and determine its dimension.
- Apply the Gram-Schmidt process to the basis you found in part (a) to construct an orthonormal basis for U.